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AI-Assisted Educational Robotics in Blended Learning:

A Study on Sustainability, Social Dimension and Ethics

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ABSTRACT

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Abstract: This thesis investigates how AI-assisted educational robotics can be used in blended learning to promote inclusion, sustainability and ethical awareness in a heterogeneous primary classroom. The study addresses a gap in how these dimensions are enacted in everyday classroom practice, especially for students with special educational needs. The research adopts a qualitative case study design involving 20 students aged 10-11, including students with special educational needs and a small group of experienced teachers who participated in a four-week robotics and AI intervention combining online and face-to-face activities. Data were generated through classroom observations, student interviews and reflective materials and were analysed thematically with a sociocultural lens focusing on interaction, participation and classroom practices. The study contributes practical insight into how AI-assisted robotics can be designed and implemented to support inclusive, values-oriented blended learning in primary education. The findings show that robotics and AI activities supported collaboration and inclusion when tasks were clearly structured and roles were defined. Sustainability-oriented practices became most meaningful when values such as fairness, resource sharing and environmental care were embedded in robot behaviours and shared classroom routines. Students engaged with AI suggestions critically rather than passively, using them as prompts for discussion about fairness, privacy and responsibility. Teachers recognised the potential of AI-assisted robotics for inclusive and socially responsible learning but reported time constraints, technical challenges and uncertainty about AI ethics, underscoring the need for further professional development. Overall, the study contributes by demonstrating both the pedagogical potential and the practical constraints of AI-assisted robotics, highlighting the need for targeted professional development on technical integration and AI ethics.

Key words: Educational Robotics, social dimension of educational robotics, social robots, distance education, AI

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1. Introduction

In 2025, the global educational robotics market reached approximately USD 1.8 billion, with projections of around USD 2.32 billion by 2026, reflecting rapid integration into classrooms worldwide (Fortune Business Insights, 2026; Future Market Insights, 2025). This growth highlights a practical challenge for education; while robots and AI-enhanced tools promise to enrich learning, they risk deepening inequalities if ethical and social dimensions such as inclusion and sustainability are overlooked (Ryalat et al., 2025). Educational robotics typically involves programmable devices that support hands-on STEM learning, often combined with AI for interactive tasks, whereas blended learning integrates these tools into a mix of in-person and online activities that can personalise and extend students' learning opportunities. In this broader landscape of digitalization, schools are expected not only to adopt new technologies but also to ensure that their use contributes to socially and sustainable forms of education (Ryalat et al., 2025; Sorayyaei et al., 2025).

Research on educational robotics has expanded quickly, demonstrating benefits for programming skills, problem-solving, motivation and engagement across different age groups and subjects. However, much of this work focuses on technical design or cognitive outcomes in face-to-face settings, paying less attention to how ethics, inclusion, sustainability and AI intersect in everyday classroom practice, especially in blended learning environments (OECD, 2024). There is limited empirical evidence on how AI-supported educational robotics affects diverse learners, including students with special educational needs, when activities span both online and in-person sessions. Questions remain about how schools can implement these technologies responsibly without reinforcing existing inequalities, creating new forms of exclusion, or neglecting environmental and social sustainability. These gaps are particularly relevant in primary education, where classrooms are heterogeneous and teachers must balance innovation with care, accessibility and practical constraints (González-Oliveras et al., 2025; Ryalat et al.,2025; Bellas et al., 2024).

Against this backdrop, this thesis focuses on how educational robotics, including modest AI-supported features, can be implemented in blended learning environments in ways that foreground inclusion, ethical awareness and social sustainability. It examines a qualitative case in one primary school, where approximately 20 students aged 10-11, including students with special educational needs and a small group of teachers with prior experience in educational robotics participate in a four-week blended intervention. The study concentrates, through the qualitative study case, on how students and teachers experience and interpret robotics activities that are deliberately designed to address social interaction, inclusion and sustainability themes and how simple AI tools are used as part of these activities; drawing on data collected through classroom observations, interviews and analysis of student work. The data were analyzed using thematic analysis to identify recurring patterns

and meanings across participants' experiences. By exploring these issues in a bounded, realistic setting, the thesis seeks to contribute to current debates on learning, digitalization and sustainability, offering practice-oriented insights into how educational robotics and AI can be used as thoughtfully designed, ethically grounded components of more inclusive and socially sustainable education. The aim and research questions guiding the study are presented in the following chapter.

This thesis contributes to existing research by examining educational robotics as a socio-technical practice embedded in the everyday routines of a heterogeneous primary classroom rather than as a short, researcher-led intervention. It extends previous work on blended robotics activities by focusing explicitly on how social interaction, inclusion and ethical awareness are enacted across online and face-to-face sessions and how AI tools participate in these processes. By bringing together students' and teachers' perspectives, the study offers empirically grounded, practice-oriented insights into how robotics and AI can be designed and facilitated to support socially sustainable participation in ordinary school conditions.

2. Aim and Research Questions

The aim of this thesis is to explore how AI-supported educational robotics can be designed and implemented within a blended learning environment to promote ethical awareness and socially sustainable learning for diverse groups of students, including those with special educational needs.

This overarching goal is purpose-driven in the sense that it does not simply describe what happens when robots enter the classroom, but seeks to understand how specific design choices, classroom practices and ethical considerations shape students' social interactions and levels of participation. By focusing on the experiences and perceptions of both students and teachers, the study aspires to illuminate how educational robotics can move beyond purely technical or cognitive outcomes and contribute to more equitable forms of technology-enhanced learning. In this study, "engagement with sustainability-related themes" refers to how students notice, talk about and participate in concrete classroom practices connected to sustainability (for example, sharing resources fairly, caring for the classroom environment, or discussing fairness and responsibility when using robotics and AI).

The study adopts an exploratory, qualitative orientation rather than a narrowly technical evaluation. It investigates how educational robotics and blended learning intersect with key themes such as inclusion, social skills, group dynamics, sustainability and emerging AI features, including modest uses of AI as support for idea generation, planning and reflection. These elements are treated as mutually shaping aspects of the same pedagogical reality: ethics, inclusion, sustainability and AI are understood as interconnected dimensions of how learning is organised and experienced in the blended robotics activities, not as separate, isolated topics.

To keep the scope realistic in relation to time, resources and context, the research focuses on one blended-learning case with a small group of students and a limited number of teachers who already have some experience with educational robotics. The intervention is bounded in time, taking place over a few weeks and uses accessible, age-appropriate tools. Within these boundaries, the study aims to produce rich, contextualised insights rather than generalisable claims.

Research Questions:

1) How does the integration of educational robotics in blended learning influence students' social interactions, collaboration and engagement with sustainability-related themes?

2) How do teachers perceive the potential and limitations of educational robotics for fostering inclusive and socially sustainable learning environments for diverse student groups, including students with special educational needs?

3) How do participants perceive ethical opportunities and risks in educational robotics activities in blended settings?

In the analysis, as we will see further below, RQ1 is approached in two closely related but analytically distinguishable strands. Interactional processes refer to how students participate, collaborate and position themselves and others during the robotics activities, whereas sustainability-oriented practices refer to how they engage with concrete issues such as sharing resources fairly, caring for the classroom environment and discussing responsibility in using robotics and AI. Treating these strands separately in the presentation of findings allows the study to trace both how students work together and how they relate their actions to broader sustainability concerns.

3. Background

3.1 Educational Robotics and digitalization

Rapid digitalization has reshaped how knowledge is accessed, shared and created in schools, with online platforms, learning management systems and networked devices becoming part of everyday classroom practice. Within this broader transformation, educational robotics has emerged as a visible strand of digital innovation, bringing programmable artefacts and physical computing into teaching and learning. Robotics activities are often presented as a way to promote twenty-first century skills such as problem-solving, creativity, collaboration and digital literacy and they are increasingly embedded in wider "smart learning" ecosystems that link robots with AI-driven tools, sensors and online environments (Adel, 2024; OECD, 2024). At the same time, policy discourses around Society 5.0 and Industry 5.0 emphasise that such technologies should support human-centred, inclusive and sustainable education rather than purely efficiency-oriented models (Carayannis & Morawska, 2022). These concerns with inclusion, ethics and sustainability are introduced here at a general level and are developed more systematically in later sections of this chapter, where they frame the analytical focus of the study.

This wider digital context creates both opportunities and challenges for schools. On one hand, digital tools and robotics kits can expand access to rich learning resources, enable flexible, blended arrangements and support more student-centred pedagogies. On the other hand, they raise questions about equity of access, the readiness of teachers and the ethical and sustainable use of technology in everyday practice. For example, not all schools can afford the same robotics platforms or infrastructure and not all students have equal access to devices or connectivity outside school. Moreover, the integration of AI-enhanced tools into robotics and blended learning introduces new issues around data protection, transparency and long-term sustainability of tools and practices (Carayannis & Morawska, 2022; Adel, 2024; OECD,2024). Against this background, the present study situates educational robotics not only as a promising digital innovation, but as a socio-technical practice that must be understood in relation to inclusion, ethics and sustainability in a concrete school context, themes which are taken up in more detail in Sections 3.3-3.4.

At the same time, dominant claims about the benefits of educational robotics are not uncontested. Several studies note that access to robotics kits, stable infrastructure and teacher training is uneven, which can reproduce or deepen existing inequities between schools and student groups and that ambitious designs sometimes clash with time constraints, assessment pressures or limited support for diverse learners in everyday classrooms (OECD,2024; Oleinikov et al.,2020). Highlighting these tensions underscores the need to study how educational robotics is implemented under realistic

conditions and how questions of equity, access and classroom organisation shape its social and ethical implications. This critical perspective directly informs the present study's focus on a blended robotics setting in an ordinary primary school, where issues of access, time and classroom organisation are treated as central analytical concerns rather than background constraints.

3.2 Educational robotics and blended learning

Educational robotics refers to the use of programmable robots and associated hardware–software systems as learning tools that enable students to design, build and program artefacts while engaging with authentic problems across subjects (Gubenko et al., 2021; Tzagkaraki et al., 2021). It can involve modular systems such as LEGO Education kits, SPIKE Prime and its successors, Arduino-compatible sets or microcontroller boards like micro:bit, which integrate sensors, actuators and often block-based programming environments to make computational thinking and engineering concepts accessible to young learners. Educational robotics is frequently described as a holistic approach that fosters scientific curiosity and engagement while promoting competencies such as initiative, responsibility, autonomy, creativity and teamwork, as well as social skills and self-esteem (Guevara et al., 2025; Tselegkaridis & Sapounidis, 2022). These activities are implemented in primary, lower secondary and higher education and have been applied across disciplines, including science, technology, arts and social studies.

Many educational robotics activities draw on constructivist and socio-constructivist traditions, including Piaget's focus on active knowledge construction, Vygotsky's emphasis on collaborative learning and Bruner's discovery learning. Studies report positive effects on critical thinking, problem-solving, metacognitive skills and basic programming knowledge (Piedade et al., 2020; Younis et al., 2023). Robotics tasks are often integrated into student-centred, project-based learning, where learners design and realise their own projects instead of merely assembling predefined models and where trial-and-error with visual programming environments such as Scratch is a central learning mechanism (Piedade et al., 2020). Educational robotics has also been shown to align with the needs of some students with specific learning difficulties; for example, the multimodal and interactive nature of robotics can support students with dyslexia, while adaptive robots allow for personalisation of movements and behaviours in ways that enhance creativity and engagement (Fung et al., 2025; Glava & Seretis, 2022). The above studies consistently report positive effects of educational robotics on collaboration, critical thinking and engagement. However, most were conducted in face-to-face settings with relatively homogeneous student groups and few examine how these benefits hold when robotics activities are distributed across online and in-person sessions in a blended format.

Distance education and blended models offer flexibility in scheduling and pace, enabling students to access materials and practise coding at their own rhythm while still benefiting from in-person interaction with teachers and peers (Kerimbayev et al., 2023; Hong, 2024). Online elements such as simulations, design tools (e.g. Tinkercad), multimedia resources and discussion forums can extend exploration and reflection beyond the classroom, while face-to-face sessions are used for building, testing and group collaboration (Tselegkaridis & Sapounidis, 2022; Silva et al., 2023; Edisherashvili et al., 2022). In the context of educational robotics, a blended approach can support continuous participation: students may sketch and test code online, then refine, debug and discuss their robotic creations in the classroom, reinforcing both technical skills and social interaction (Silva et al., 2023; Rasheed et al., 2020). These studies suggest that blended formats do not simply add online components to robotics work, but reconfigure when and how students collaborate, negotiate roles and sustain participation across physically and digitally mediated spaces.

Several studies highlight that blended robotics environments can cultivate twenty-first-century skills such as collaboration, communication and creativity. Virtual group projects and online problem-solving challenges can be combined with physical robot work to support teamwork and digital communication skills (Rasheed et al., 2020; Kerimbayev et al., 2023). Within this broader shift, educational robotics has been discussed as part of emerging "smart" learning environments that link AI, connected devices and physical artefacts, moving from Industry 4.0's emphasis on automation and efficiency towards more human-centred and socially responsive visions associated with Industry 5.0 (Adel, 2024). While the literature on blended learning and the literature on educational robotics have each grown substantially, they have developed largely in parallel: studies on blended learning rarely focus on robotics as the learning activity and studies on educational robotics rarely investigate blended delivery models. This structural separation between two research traditions, one on educational robotics and one on blended learning, leaves a gap in understanding how the social and collaborative dynamics of robotics work change when activities are split across online preparation and face-to-face building sessions. Building on this gap, the present study asks how blended robotics activities can be designed not only for cognitive outcomes, but also for inclusion, social interaction and sustainability.

3.3 Social dimension, inclusion and AI in educational robotics

Beyond cognitive and technical outcomes, a growing body of work examines the social dimension of educational robotics. This field is interdisciplinary, integrating pedagogical, psychological and technological perspectives. It is useful to distinguish between social robots, robot-assisted learning and the social dimension of educational robotics: social robots are designed to communicate with

humans in socially understandable ways; robot-assisted learning focuses on robots as cognitive tools to support specific curricular goals; and the social dimension of educational robotics examines how these technologies influence interpersonal relationships, social skills and students' inclusion in the educational process (Pnevmatikos et al., 2022; Fachantidis et al., 2020; González-Oliveras et al., 2025).

Moreover, robotic activities can encourage students to interact, collaborate and communicate as they jointly design and program robots. Working in groups, learners exchange ideas, support each other and coordinate their actions to solve challenges, which can strengthen teamwork and peer relationships (González-Oliveras et al., 2025; Fachantidis & Triantafyllidou, 2022). Experiential robotics tasks may be especially supportive for students who struggle to participate in more traditional whole-class discussions, offering them a concrete object of focus and a structured framework for participation. Studies on storytelling with robots show that children's emotional engagement depends not only on story content but also on the robot's "personality" and expressive behaviour, suggesting that robots can function as socially active agents within learning experiences (Velentza, Pliasa, & Fachantidis, 2021).

A particularly important line of research concerns the use of socially assistive robots (SARs) to support children with Autism Spectrum Disorder (ASD). Findings indicate that SARs can have positive effects on social participation and communication, both in individual and small-group contexts (Schiavo et al., 2024). For example, the robot Daisy has been used to scaffold triadic interactions among a child with ASD, a neurotypical peer and the robot, enhancing eye contact, joint attention and verbal interaction (Fachantidis et al., 2020). More broadly, experiential, hands-on robotics activities can sometimes lead to changes in participation patterns, with students taking more active roles in group work, expressing ideas and assuming responsibilities within teams (Ponticorvo et al., 2020). Combinations of social robots with other technologies, such as augmented reality, have also been explored to support collaborative inquiry and spatial understanding in subjects like geography, while simultaneously fostering social skills (Pasalidou & Fachantidis, 2022; Pasalidou et al., 2023). Although these findings are encouraging, a common limitation across this body of work is that most studies rely on short, researcher-controlled interventions rather than naturalistic classroom settings and they tend to focus on individual social behaviours (e.g., eye contact, turn-taking) rather than on how inclusion is experienced by the students themselves within everyday group work (Fachantidis et al., 2020; Pasalidou & Fachantidis, 2022).

In this study, the emerging use of AI is understood as part of this social dimension: simple, classroom-oriented AI tools become additional mediating artefacts that shape how students

coordinate, reflect and negotiate inclusion during robotics activities. In parallel, AI has entered educational technologies in multiple ways, including adaptive learning systems, recommendation engines and intelligent tutoring. AI can enhance personalisation and provide data-driven insights that help institutions refine curricula and support students (De la Vall & Araya, 2023; Wang et al., 2024; Eden et al., 2024). However, the integration of AI into robotics and blended learning also raises ethical questions about privacy, transparency, bias and power relations.

In the present study, based on the above, AI is understood in a narrow, classroom-oriented sense; simple tools that generate ideas, support planning or prompt reflection, rather than complex autonomous systems or advanced learning analytics. These tools are embedded in the robotics project primarily to support collaboration and reflection and to stimulate discussion about fairness, inclusion and responsibility. For instance, AI may be used to propose possible behaviours or dialogue lines for a "helper" robot, suggest steps for group planning or offer reflection questions at the end of a session. Such interactions are mediated by the teacher, who selects and adapts prompts and emphasises that AI outputs are suggestions to be critically evaluated, not instructions to be followed automatically. Notably, few investigate how students and teachers actually perceive and negotiate AI-generated suggestions in real classroom situations, especially when ethical questions about fairness, bias or data use are at stake. This imbalance between what AI can theoretically offer and what is empirically known about its classroom reception further motivates the present study's focus on participants' lived experiences. This limited and transparent use is intended to keep AI age-appropriate for primary students while aligning with a human-centred, ethically aware approach to emerging technologies.

In addition to these general developments, recent work on AI in education highlights the importance of framing AI tools as objects for critical engagement rather than as neutral technical supports. Simple, classroom-oriented AI systems can be used to surface questions about fairness, bias and responsibility by inviting students and teachers to reflect on why a particular suggestion is appropriate or not and whose perspectives are represented in AI-generated outputs. In the context of educational robotics, this means that AI can function not only as a generator of ideas or explanations, but also as a prompt for discussion about how robots should behave toward classmates, how data about students are used and what counts as "respectful" or "sustainable" behaviour in a group (Wang et al., 2024; Eden et al., 2024). In this study, AI is therefore positioned simultaneously as a practical scaffolding for planning and reflection and as a socio-technical element that helps students and teachers articulate and negotiate ethical and social sustainability questions in a concrete, age-appropriate way.

3.4 Sustainability, ethics and the research gap

Digitalization and the growing presence of AI and robotics in schools need to be considered in relation to sustainability and ethics. Sustainability in education is often framed through the Sustainable Development Goals (SDGs) and educational robotics has been linked in particular to SDG 4, which calls for inclusive and equitable quality education and promotes lifelong learning opportunities for all. Sub-goals 4.a and 4.c emphasise upgrading education facilities to be child- and disability-sensitive and providing safe, effective learning environments, as well as strengthening the capacity of teachers (Schina et al., 2020). From a social sustainability perspective, the use of educational robotics raises questions of equitable access, responsible technology use and the creation of inclusive learning environments. Educators are challenged to implement robotics in ways that uphold fairness, empathy and social responsibility and to consider how technological choices intersect with issues such as disability, language, socio-economic status and local school resources (Rumaiza, 2025; Lindberg & Haglind, 2021).

In this sense, sustainability in educational robotics goes beyond environmental concerns to include long-term pedagogical and ethical considerations. Rapid technological development means that tools and platforms may change or become obsolete, requiring teachers to design activities that are robust to such changes and can be adapted without disadvantaging particular learners. At the same time, digital tools used in robotics and blended learning collect and process student data, which raises ethical issues related to privacy, consent and data governance. Teachers must navigate these challenges while also attending to classroom dynamics, ensuring that the introduction of robotics and AI supports rather than undermines students' wellbeing and sense of safety. This includes protecting personal data, following ethics guidelines and designing activities that allow all students, including those with special educational needs, to participate meaningfully and safely (Lindberg & Haglind, 2021; Carayannis & Morawska, 2022).

Earlier research thus shows that educational robotics and blended learning can foster important skills and engagement; that robotics activities have a significant social dimension with potential benefits for inclusion, especially for students with ASD; and that digitalization and AI in education are increasingly discussed in relation to sustainability and ethics. However, several gaps remain. Much of the existing literature either focuses on technical aspects and cognitive outcomes in face-to-face settings or explores social robots in highly structured interventions outside everyday classroom practice (Alimisis, 2020; Piedade et al., 2020; OECD,2024). There are fewer empirical studies examining how modest, AI-supported robotics projects are implemented in real primary classrooms that combine online and in-person activities and how such implementations shape everyday social interactions, inclusion of students with special educational needs and engagement with concrete sustainability-related classroom practices. In particular, little is known about how teachers and

students themselves perceive the ethical opportunities and risks associated with these technologies in blended settings and how these perceptions relate to data practices, power relations and student vulnerability. Empirically, sustainability in this study is traced through students' talk and actions around concrete classroom practices, such as how they distribute roles and resources, how they care for shared materials and how they discuss fairness and responsibility in relation to the robots and AI prompts (OECD,2024; Oleinikov et al.,2020). Teachers' reflective notes are also examined for how they frame long-term pedagogical and ethical considerations, including which students are enabled or hindered from participating in socially sustainable ways.

Addressing this gap is important for both research and practice. For educators and school leaders, there is a growing need for context-sensitive evidence on how to integrate educational robotics and simple AI tools in ways that support participation, social sustainability and student wellbeing. For policy makers and curriculum designers, insights into the ethical and social consequences of robotics-enhanced blended learning can inform frameworks and guidelines aligned with inclusion and sustainability agendas such as SDGs.

3.5 Sociocultural Theory

Vygotsky's sociocultural theory views learning as a fundamentally social process that happens first between people and only later within the individual. Rather than seeing knowledge as something transmitted from teacher to student, Vygotsky argued that children construct understanding through interaction with more knowledgeable others, such as teachers, peers, adults and with the cultural tools available in their environment, such as language, symbols and artefacts. A central idea is that cognitive development cannot be separated from the social and cultural context in which it occurs; the ways children think, speak and act in learning situations reflect the practices and tools of their communities (Vygotsky, 1978; Allman, 2023). In educational settings, this means that classroom tasks, group arrangements and technologies are not neutral; they actively shape what and how students learn, especially in relation to robotics and AI.

One of the most influential concepts in this theory is the Zone of Proximal Development (ZPD), defined as the distance between what a learner can do independently and what they can do with appropriate guidance and support. Within the ZPD, "scaffolding" is provided by more capable peers or adults who offer temporary, adjustable support, such as hints, examples, questions or demonstrations, that enables the learner to tackle tasks that would otherwise be out of reach. As competence grows, the scaffolding can be gradually removed so that the learner takes increasing responsibility. Although the term scaffolding was introduced later by researchers inspired by Vygotsky rather than by Vygotsky himself, it captures the dynamic, contingent support that

characterises learning in the ZPD (Shabani et al., 2010; Verenikina, 2003; Wan Hamedi et al., 2025; Allman, 2023). This view of learning highlights the importance of collaborative work, guided practice and carefully designed tasks that are neither too easy nor too difficult. It also implies that tools and technologies can participate in scaffolding, for example by breaking down tasks, giving feedback or structuring sequences of action, usually in ways that complement rather than replace teacher and peer support.

Another key concept is mediation, which refers to the role of cultural tools and signs in shaping human activity and thought. For Vygotsky, people rarely act directly on the world; instead, they use mediating artefacts such as language, diagrams, numbers or physical tools that transform both the activity and the learner. In contemporary classrooms, digital technologies, including robots, programming environments and AI systems, can be seen as powerful mediating tools. They can structure interaction (for example, by requiring students to coordinate steps in a program), make thinking visible (through code or sensor readings) and create new forms of collaboration around shared artefacts. From a sociocultural perspective, analysing how students use and talk around these tools provides insight into their learning processes, the social organisation of activity and the opportunities that different designs create for participation and inclusion (Daniels, 2008; Allman et al, 2023; Bildik & Altun, 2022).

4. Method

4.1 Research design

This study adopts a qualitative case study design focused on a single blended-learning setting where AI-supported educational robotics activities are implemented with a small group of students and their teachers. A case study is appropriate because it allows an in-depth investigation of a contemporary phenomenon within its real-life context, using multiple data sources and a clearly bounded case, one class, one school, one four-week project (Yin, 2018). A single, information-rich case is sufficient here because the aim is not statistical generalisation across many settings, but analytic generalisation: developing a theoretically informed account of how AI-supported robotics can mediate inclusion, interaction, ethics and sustainability in depth in one context that can then be compared to other cases in future work. The approach is grounded in an interpretivist paradigm, seeking to understand how participants experience and make sense of inclusion, social interaction, ethics and sustainability in their specific educational context and to produce rich, context-dependent insights rather than generalisable causal claims. This interpretivist stance also informs the analytical choices outlined later in the chapter, where concepts such as mediation, scaffolding and participation are used to interpret participants' situated meaning-making rather than to test predefined hypotheses (Pervin & Mokhtar, 2022).

A structured review of recent literature on educational robotics, blended learning, inclusion, ethics, sustainability and AI in education**,** informs the conceptual framing of the case, the formulation of the research questions and the analytic focus. Rather than providing a comprehensive, standalone review, this literature is used selectively to identify specific tensions and gaps that the present case study is designed to explore. To reduce the risk of confirmation bias, the review deliberately includes studies that report both benefits and potential challenges of educational robotics and AI and the analytic process remains open to empirical patterns that complicate or contradict expectations derived from prior research.

4.1.1 Sociocultural Theory in the thesis

Vygotsky's sociocultural theory is used in this thesis as a lens for understanding how learning and social interaction unfold around the robots and AI tools in the blended activities. Central concepts are the Zone of Proximal Development (ZPD) and scaffolding, which describe how learners can achieve more in collaboration with others and with mediating tools than they could alone (Allman, 2023). In the context of educational robotics, the robot, the block-based programming environment and the simple AI tool, function as mediating artefacts that structure students' talk, joint problem-solving and roles within the group; while teachers and more experienced peers provide scaffolding through

prompts, questions and demonstrations. This perspective aligns closely with the study's focus on group work, social interaction and how students with different needs, including those with ASD, participate in shared robotics tasks. It also supports a strengths-oriented view of students' participation, emphasising what learners can do with appropriate support and meaningful tools rather than framing them primarily in terms of individual deficits or difficulties.

This theoretical lens directly informs the formulation of RQ1, which asks how the integration of educational robotics in blended learning influences students' social interaction, collaboration, sense of inclusion and engagement with sustainability-related practices. From a sociocultural standpoint, these processes are not just individual traits but are co-constructed through interaction in specific activities and with specific tools. The design of the Helper Bot and Sustainability Bot tasks deliberately creates opportunities for students to work within each other's ZPD: mixed-ability groups allow peers to model behaviours, explain code and support turn-taking, while the teacher adjusts the difficulty of tasks, provides hints and uses questions to draw quieter students into the activity. Observations therefore pay particular attention to how scaffolding is distributed (who helps whom, how and when) and to the ways in which the robot and AI prompts open or close space for different students to contribute.

RQ2, which focuses on teachers' perceptions of the potential and limitations of educational robotics for fostering inclusive and socially sustainable learning environments, is also grounded in this sociocultural view. In Vygotskian terms, teachers are key mediators who design the learning environment, choose and adapt tools and orchestrate interactions so that students can work in their ZPDs. When teachers reflect on which students were able to participate, which scaffolds were effective or insufficient and how the blended format shaped collaboration, they are implicitly commenting on how well the sociocultural conditions for learning were created. The reflective teacher notes therefore capture not only practical impressions (e.g., time constraints, technical issues) but also teachers' judgements.

Finally, RQ3 which explores how participants perceive ethical opportunities and risks when AI and robotics are integrated into blended learning, can be read through a sociocultural lens that emphasises power, voice and participation. From this perspective, ethical issues are not only about abstract principles but also about how technologies mediate relationships and who has control over tools and decisions in the learning activity. Interview questions that ask students whether they felt listened to, whether AI suggestions were fair or helpful and whether the robot supported everyone or only some classmates, invite them to comment on how mediation and scaffolding were enacted in practice. Teachers' reflections on moments when AI prompts confused or empowered students, or when certain

learners were left out of group work, similarly provide insight into how technological and social mediation intersect. This focus on who can act, speak and influence decisions connects the study to broader sociocultural debates on agency and control in technology-mediated learning, where tools are understood as shaping and sometimes constraining, possibilities for participation. In this way, Vygotsky's sociocultural framework helps to connect everyday interactional details with broader concerns about inclusion, agency and ethical use of AI in the classroom.

The sociocultural framework, also, shapes the interview questions used with students. Rather than asking only whether they "liked" the robot, the interviews focus on how students experienced interacting with peers, the robot and the AI tools, as well as how they felt as individual learners. For example, questions such as "Can you describe a moment when collaboration worked well and a moment when it was more difficult?", "Did you feel included and listened to in your group during the activities? What helped you feel included, or what made it harder?", or "Do you think using robotics and AI in this way is good for all students, including those who may be shy or have different learning needs?" explicitly attend to participation, mutual help and recognition in the group. Other questions (e.g., "When you think about the robotics project, what stands out most in your mind?" or "Were there any moments when you felt worried or unsure about the robot or the AI?") invite students' individual experiences and emotions. In this way, the Vygotskian focus on mediated, collaborative learning and guided participation structures not only what the study looks for in observations but also how the activities are designed and how students' experiences, both relational and individual, are interpreted in relation to the research questions. Questions 2-4 and 6-8 primarily address interactional aspects of learning (collaboration, inclusion, respect, participation), whereas questions 1, 5, 9 and 10 focus more on students' individual experiences, emotions and reflections

In analysing the empirical material, three sociocultural concepts are particularly central; mediation, scaffolding and participation. Mediation guides attention to how robots, programming environments and AI tools structure interaction and meaning-making. Scaffolding focuses the analysis on how support is distributed among teachers and peers over time and participation directs the lens to who is included, how roles are negotiated and how students with different needs position themselves within the blended robotics activities. These three concepts were prioritised because they speak most directly to the design of the blended robotics activities, the observable interactional data and the study's concern with how students are supported to take part, whereas other sociocultural notions such as internalisation or activity remain more in the background as overarching assumptions.

4.2 Context and participants

The case study is situated in one primary school, in a class of approximately 20 students aged 10-11 years. The class includes learners with diverse academic and social backgrounds, among them some formally diagnosed with mild learning or social difficulties and at least two with Autism Spectrum Disorder (ASD). Four teachers who already have experience with educational robotics and blended learning participate by implementing the activities and contributing reflective notes, but they are not interviewed.

Sampling is purposive: this class and these teachers were selected because they are already engaged with educational robotics and because the presence of diverse learners makes it possible to explore questions of inclusion, ethics and sustainability in depth. Such purposive selection of an "information-rich" case is consistent with qualitative case study traditions, where the aim is depth of understanding rather than statistical representativeness (Yin, 2018).

4.3 Blended implementation

The empirical intervention consists of a four-week educational robotics project implemented in a blended learning format. Students work in small, mixed-ability groups with robotics kits (for example LEGO SPIKE or comparable systems) and block-based programming environments. Online sessions are used primarily for planning, simple coding tasks and reflection, while face-to-face sessions focus on building, testing and observing group interaction around the tools. This distribution is pedagogically motivated by the aim to reserve in-person time for hands-on collaboration and social participation, while using online time for more individual or small-group preparation and reflection.

4.4 Data collection methods

To address the research questions, data are collected from two main sources, complemented by the researcher's field notes:

- **Classroom observations and researcher field notes** (RQ1 and supporting RQ3): During face-to-face robotics sessions, the researcher conducted non-participant observations focused on group interaction, role distribution, participation of students with special educational needs and references to sustainability-related practices such as sharing materials or caring for the classroom environment. A non-participant role was chosen to minimise disruption of existing classroom dynamics and to allow teachers and students to interact as naturally as possible during the activities (Mills, Gay, & Airasian, 2017). The participating teachers were responsible for planning and leading the lessons, while I remained in the classroom as an observer. Brief notes were taken during sessions and expanded into fuller field notes immediately afterwards, including reflections on ethical or AI-related moments.

- **Reflective teacher notes** (primarily RQ2 and RQ3, supporting RQ1): Participating teachers keep short, structured notes after selected sessions, documenting notable social behaviours (for example, clear moments of inclusion or exclusion, shifts in participation, conflicts, or peer support), perceived changes in inclusion and any ethical or practical issues they notice in relation to robotics, AI use and sustainability themes. The notes are guided by a simple template with prompts, for example, "moments of inclusion or exclusion," "student reactions to AI suggestions," "links to sustainability practices", to support comparability across teachers while still leaving space for open comments.

- **Semi-structured student interviews** (RQ1 and RQ3): At the end of the project, semi-structured interviews are conducted with a small number of students, including students with ASD and others representing different participation patterns. The interviews explore how they experienced working with the robot in groups, how included and listened to they felt, how they perceived the use of AI tools and whether they noticed moments related to fairness, respect or sustainability in the activities. Semi-structured interviews are chosen because they offer a balance between comparability across participants and flexibility to follow up on important issues in children's own words (Mills, Gay, & Airasian, 2017). For some students, an adapted, easy-to-understand version of the interview guide is used. Adaptations were based on teachers' knowledge of students' support needs, focusing on language simplification and pacing rather than changing the core questions.

Research Questions and data collection

- **RQ1**: This question is addressed by combining classroom observations and field notes, which capture real-time group dynamics, role distribution, peer support and references to sustainability practices; teacher reflective notes, which provide an additional perspective on how social interaction happened across sessions; and student interviews, which give access to students' own interpretations of how included and listened to they felt. Triangulating these three sources makes it possible to compare what is observed in practice with how participants themselves understand and describe their experiences, thereby strengthening the analysis of social and sustainability-related processes.
- **RQ2:** This question is addressed through structured reflective notes that teachers write after selected sessions. The notes are designed to capture not only factual observations (e.g. who participated, what difficulties arose) but also teachers' interpretive reflections: how they perceive the effect of the robotics activities on inclusion, what they see as enabling or constraining factors for socially sustainable pedagogy and how they assess the role of AI tools

in shaping group dynamics. By analysing patterns and recurring judgements across multiple notes and multiple teachers, the study can identify shared perceptions as well as points of divergence.

- **RQ3:** This question draws on all three data sources, because ethical perceptions may surface in different ways and at different moments. Student interviews are the primary source, exploring whether children felt worried and whether they perceived the tools as helpful or intrusive. Teacher notes complement this by documenting any ethical concerns that teachers noticed during the activities, such as unequal access, data-related worries or moments when AI suggestions were perceived as inappropriate or biased. The researcher's own field notes add a reflexive layer, recording situational details and power dynamics that participants may not articulate explicitly. Analysing these sources together allows the study to construct a multi-perspective account of ethical awareness, capturing not only what participants say about ethics but also what emerges implicitly through their actions and interactions.

4.5 Pilot study

Before the main implementation, a small pilot was conducted to check the feasibility and clarity of the research procedures. The pilot included one trial run of the robotics and AI lesson in class and one individual interview with a student. The aim was to test the lesson flow, the comprehensibility of the instructions and interview questions and the overall timing and organisation. The pilot indicated that the core structure of the lesson and interview guide functioned well in the school context, so only minor adjustments to wording and instructions were made; no substantial changes to the design, activities or analytic focus were required. Therefore, the pilot study functions as an initial, small-scale implementation of the same design used in the main study. Because the procedures were substantively identical to those used in the main study and changes were limited to small clarifications of phrasing, the pilot data are included in the overall dataset and analysed together with the main implementation. The pilot therefore provided both an opportunity to refine the procedures and additional assurance that the activities and instruments were appropriate, understandable and manageable within the school context.

4.6 Data analysis

All teacher notes, observation records and student interviews are compiled into text documents. The analysis follows a reflexive thematic analysis approach inspired by Braun and Clarke's model, adapted to the aims and scale of this study (Braun & Clarke, 2022). Coding is carried out manually in order to stay close to the data and to allow flexible refinement of the coding scheme. The concrete steps are:

1. Familiarisation: repeated reading of observation notes, teacher notes and interview transcripts to gain an overall sense of the material and to note initial ideas related to inclusion, social interaction, sustainability and ethics.

2. Initial coding: systematic coding of relevant segments of text, using both data-driven and concept-driven codes (e.g. “turn-taking support,” “feeling listened to,” “AI suggestion helpful/problematic,” “sustainability framed as fairness”).

3. Developing themes: grouping related codes into preliminary themes aligned with, but not restricted to, the three research questions, such as "patterns of inclusion and exclusion in group work", "teachers' reflections on constraints and opportunities", or "AI as support vs AI as risk".

4. Reviewing and refining themes: checking themes against the coded extracts and the full data set, merging or splitting themes where necessary to ensure coherence and distinction.

5. Interpreting themes: relating the final themes back to the research questions and to the conceptual background on educational robotics, blended learning, inclusion, sustainability and AI. Throughout the analytic process, reflexivity is maintained through analytic memos, coding decisions and theme revisions and regular reflection on how the researcher's positionality and assumptions about inclusion, robotics and AI may shape interpretation.

Through the reflexive thematic analysis, three overarching themes that cut across the different data sources and speak directly to the research questions. The first theme, interactional processes of collaboration and inclusion, captures patterns in how students participated, collaborated and positioned themselves and others during the robotics and AI activities. The second theme, sustainability-oriented practices and classroom responsibility, focuses on how students engaged with issues such as sharing resources fairly, caring for the classroom environment and negotiating responsibility in relation to robotics and AI. The third theme, ethical engagement with AI and robotics in practice, traces how students and teachers made sense of AI suggestions and robotic behaviours in terms of fairness, privacy and care. It also, includes moments of curiosity, negotiation and concern. Each theme integrates interviews, observation and teacher note material.

4.7 Reliability and Validity

The reliability and validity of the study is evaluated through the criteria of credibility, transferability, dependability and confirmability (McGrath et al.,2019; Mills et al, 2017). Credibility refers to the extent to which the findings accurately represent participants' experiences and it is strengthened

through triangulation of data sources, classroom observations, teacher notes and student interviews, so that no single account dominates the analysis. Member checking, whereby preliminary findings or interpretations are shared with participants to verify accuracy and completeness, is an additional strategy for enhancing credibility (Mills et al., 2017; Stahl & King, 2020). In this study, emerging interpretations are discussed informally with participating teachers, who can confirm, challenge or add nuance to the researcher's reading of classroom events; this form of member checking focuses on teachers as key informants and does not involve student participants directly, which also reflects ethical considerations about placing additional demands on children. Dependability and confirmability are supported by maintaining a transparent audit trail of analytic decisions, including coding logs, theme-development notes and reflexive entries documenting the researcher's assumptions and evolving role throughout the project. Because the study relies on a single, bounded case, claims to transferability are modest; however, the provision of rich, detailed descriptions of the context, participants and activities enables readers to judge the relevance of the findings to their own settings (Mills et al., 2017; Yin, 2018).

As a researcher, I enter the field with prior experience and interest in digital technologies in education, including educational robotics and AI-supported tools. This background provides familiarity with the technical and pedagogical possibilities of such tools, but it also risks predisposing me to view them as inherently beneficial. To mitigate this, I consciously attend to moments of tension, resistance and difficulty in the data and I make explicit in the analysis when my interpretations are informed by my own assumptions rather than by participants' accounts. I also reflect on my position in the classroom, as an external observer collaborating with teachers and how this may have influenced what students and teachers chose to share or emphasise during observations and interviews. This reflexive stance aligns with the interpretivist paradigm underpinning the study, which views knowledge as co-constructed and treats the researcher's perspectives as resources to be critically examined.

4.8 Methodological limitations and challenges

As a small, purposively selected single-case study, the findings cannot be generalised to all schools, age groups or national contexts (Yin, 2018). The four-week duration may not capture longer-term changes in social interaction, inclusion or attitudes toward educational robotics and AI. Reliance on observations, teacher notes and student narratives means that the results are shaped by participants' accounts and the researcher's interpretations, despite efforts at reflexivity and triangulation. The blended format may introduce variability in participation, for example if students engage differently online and in face-to-face sessions or if technical problems disrupt activities.

Differences in students' prior experience with robotics, time constraints within the school timetable and potential sensory overload for some students with ASD (e.g. noise, lights or movement) present additional practical challenges. Where possible, these challenges were managed through small practical adjustments (for example, flexible groupings, brief breaks and reducing sensory load) and ongoing coordination with teachers. These constraints are acknowledged so that the scope and claims of the study remain transparent and appropriately modest.

4.9 Ethical considerations

The study follows national and institutional ethical guidelines for research with children in school settings, including principles of informed consent, assent, confidentiality and the right to withdraw (European Commission, 2021; Swedish Research Council, 2024). In line with the Swedish Research Council's guidance and the Swedish Ethical Review Act, research involving children is planned and conducted so that risks are minimised, human dignity is respected and informed consent and data protection requirements are met (Swedish Research Council, 2024).

Written informed consent is obtained from parents or legal guardians and age-appropriate assent procedures are used with students, ensuring that they understand in simple language what participation involves and that it is voluntary. When assessing children's capacity to understand, information is adapted in complexity and language to their developmental level and children's wishes not to participate are respected even when parental consent has been granted, in line with Swedish guidance on consent for minors in research. For students with ASD or other special educational needs, information about the study is provided in simplified language and, where helpful, supported with visual aids or additional explanation in collaboration with teachers or special education staff, in line with recommendations for ethical autism research (Fachantidis et al., 2020; Swedish Research Council, 2024).

Confidentiality is ensured by assigning pseudonyms, removing identifying details from transcripts and notes and storing digital files securely with access restricted to the researcher (Swedish Research Council, 2024; National and Kapodistrian University of Athens, n.d.). Any audio recordings of interviews or observations are handled according to institutional data protection procedures and are not shared beyond the research team, with particular attention to the safeguards required when processing personal data about children and potentially sensitive educational information. Particular care is taken not to disclose identifying information about students with special educational needs in the reporting of results, in order to avoid stigma or unintended disclosure and to uphold the protection interest emphasised in Swedish guidance on openness and confidentiality in research (Swedish Research Council, 2024).

Power dynamics are addressed by emphasising that participation is voluntary and that students can decline or withdraw at any time without negative consequences and by making clear to teachers that the research is not an evaluation of their performance (May & Perry, 2022). Throughout the intervention, teachers act as partners in monitoring student wellbeing; tasks and environments can be adapted if signs of distress or overstimulation appear, especially for students with ASD, reflecting the Swedish Research Council's emphasis on balancing research value with potential risks and on giving particular protection to vulnerable groups. This also reflects broader principles in research with children that stress the need to balance research aims with the safety and wellbeing of participants and to minimise the risk of emotional or psychological distress (Swedish Research Council, 2024).

Because the research is conducted under the auspices of a Swedish higher education institution while data collection takes place in another European country, ethical responsibility is treated as multi-layered. The study is therefore designed to comply simultaneously with Swedish national research ethics standards and with the national ethical regulations and legal requirements of the host country, Greece, particularly in relation to research with children, consent procedures and data protection. This dual alignment strengthens the ethical rigor and contextual validity of the study and clarifies how legal, institutional and EU-level frameworks jointly shape the protection of participating children.

4.10 Ethical tensions in AI-supported educational robotics research with children

Research that combines AI, educational robotics and children with ASD raises several interconnected ethical tensions that are directly relevant to this study. First, the collection of interaction data by AI-enhanced robots creates privacy risks, since children, especially those with disabilities, may not fully understand what is being recorded, stored or analysed and their capacity to give meaningful consent is limited (Kurian, 2023; Kurian 2025). Second, AI systems trained on narrow or biased datasets risk reproducing ableist assumptions, for example by treating atypical communication styles as deficits rather than differences, which can lead to unfair or stigmatising robot behaviours (Singh et al., 2022). Third, children with ASD may develop strong emotional attachments to robots, raising questions about trust, deception and dependency, particularly when a robot's socially engaging behaviour is designed to elicit responses that the child may interpret as genuine friendship or care (Kouroupa et al., 2022). Leading studies address these tensions by adopting participatory design approaches that include autistic children and their caregivers in the development process, by applying child-rights frameworks such as the UNICEF Guidelines on AI for Children and by building in opt-out mechanisms, sensory accommodations and continuous ethical oversight throughout the intervention (Lemaignan et al., 2021; Singh et al., 2022).

In the present study, these concerns are mitigated by using AI only in limited, teacher-mediated interactions where no personal student data are collected or processed by the AI tools. The robotics kits and AI features are used as pedagogical supports for brainstorming and reflection, not as autonomous data-gathering systems. Activities are designed with built-in flexibility, including sensory accommodations, adaptable tasks and voluntary participation, so that the wellbeing and dignity of all students and especially those with ASD, are safeguarded throughout the project. In this study, these ethical considerations directly shape the analytical focus of RQ3. RQ3 is investigated with explicit attention to how privacy risks, potential algorithmic bias and the possibility of emotional attachment to robots influence what can be understood as ethical or unethical practice in the classroom context. Accordingly, episodes of interaction and participants' accounts are interpreted with these ethical tensions in mind, so that they become central to answering RQ3 rather than remaining background concerns. Together with the trustworthiness strategies, ethical protocols and methodological transparency described above, these measures aim to ensure that the study is conducted in a manner that is not only scientifically rigorous but also socially responsible.

5. Results

5.1 Overview of data sources and core activities

This chapter presents the findings in relation to the three research questions by organising the analysis around three key themes that emerged from the thematic analysis: The first theme, Interactional processes of collaboration and inclusion, the second theme, Sustainability-oriented practices and classroom responsibility and the third theme, Ethical engagement with AI and robotics in practice. Each theme integrates evidence from student interviews, classroom observations (face to face and online) and teacher reflections before, during and after the intervention. This thematic structure allows patterns across data sources to be examined in a coherent and complementary way.

Two core activities form the backbone of the project; a "Helper Bot" task that focuses on classroom support and inclusion and a "Sustainability and Inclusion Challenge Bot" task that explicitly connects robotics work to sustainability-related classroom practices and inclusive group participation. In both activities, simple AI-based tools are used in a limited, teacher-mediated way, for example to generate ideas for robot behaviours, suggest prompts or support reflection on how the robot should act towards classmates. Thus, teachers retain control over when and how AI input is taken up in the classroom in line with the ethical considerations. These activities provide a pedagogical context within which the findings on social interaction, inclusion, ethics and sustainability presented in this chapter were generated.

5.2 Interactional Processes of Collaboration and inclusion (RQ1 and RQ2)

The first theme, captures how students engaged with one another during the robotics and AI activities, how participation was distributed within groups and how these interactional patterns were shaped by teacher support. Across student interviews, classroom observations and teacher reflections, collaboration was consistently described as one of the most distinctive and motivating aspects of the project. Students often contrasted the activities with more conventional classroom lessons, describing them as "*fun*", "*exciting*", "*interesting*" and "*different from normal lessons*." These descriptions were closely connected to the opportunity to work actively with peers, test ideas and interact directly with the robot.

For many students, participation was sustained by the practical nature of the tasks. The possibility of experimenting, making mistakes and immediately trying again appeared to foster both engagement and persistence. As students explained, they enjoyed "*trying again when it did not work*" and wanted to "*see if we can make the robot do something new*." Several students, also, noted that they looked forward to these sessions more than ordinary lessons. A key condition for participation was the

distribution of meaningful roles. Students referred to roles such as “coder”, “builder” and “tester” and several indicated that these roles helped them remain engaged throughout the activity. One student stated, “*I was the tester, so I always had something important to do even if I wasn’t coding*.” Students felt included when they could identify a clear responsibility within the group.

Students, also, described collaboration as supportive and productive when the work was organised in ways that allowed everyone to contribute. One student explained, “*It was easier in a group because when I didn’t know what to do, someone else explained it and we fixed the robot together*.” Another contrasted the experience with routine lessons by saying, “*In normal lessons I sometimes get bored, but with the robot I wanted to try again when it didn’t work and my group waited for me to finish*.”

Observation data reinforced these accounts. In groups where roles were explicitly assigned, participation was generally more balanced and students actively monitored one another’s involvement. For example, students were heard reminding peers that “*now it is your turn to test*”. Role rotation over time further supported engagement, as it enabled students to move between functions and experience different aspects of the task. By contrast, in groups where roles were not clearly structured, participation was more uneven. Some students dominated access to the robot or tablet, while others remained on the margins, “just watched,” or waited passively for a turn that did not always come.

When students voluntarily swapped roles, so that everyone could experience different aspects of the activity, this flexibility seemed to increase engagement and allowed students to discover new strengths. In interviews, students mentioned that they were surprised to find that they enjoyed testing or troubleshooting more than they had expected. In the online components, students were observed working more individually when drafting ideas or experimenting with simple code, but still used chat and breakout rooms to ask questions, share screens and show solutions to peers.

Teacher reflections closely aligned with these observed patterns. At the beginning of the intervention, teachers described group work as uneven, with some students taking control and others remaining passive. Over time, however, they observed greater willingness among students to collaborate, experiment and persist when tasks became difficult. Teachers noted that students who were usually quiet in whole-class settings became more active in small-group work, particularly when the task involved a specific, manageable responsibility. They also described the emergence of peer learning, with students explaining ideas such as loops or sensors to one another and offering encouragement after failed attempts.

At the same time, the data show that collaborative participation was not always inclusive or balanced. Students described disagreements, unequal access to the robot and moments when their ideas were dismissed or ignored. Some spoke explicitly about exclusion or hesitation, noting, "*I wanted to do more but I couldn't participate*", "*I was…shy and I didn't want to make a mistake*." These comments point to the emotional and social dimensions of participation, showing that students' involvement was shaped not only by interest but also by confidence, group climate and the perceived safety of contributing. Observations similarly showed that quieter students could remain in the background unless structures were in place to support turn-taking and inclusion.

Teachers responded to these challenges by introducing role rotation, closely monitoring group interactions and establishing explicit rules for sharing access to the robot and related materials. They noted that when groups were composed with attention to diversity of skills and personalities and when roles were clearly defined, collaboration tended to be more constructive. Conversely, groups composed without such considerations were more prone to conflict or to uneven participation. Teachers thus saw group composition and role structuring as essential components of inclusive robotics and AI pedagogy.

In the follow up reflections one month after the project, teachers reported that, overall, they had observed more balanced participation in robotics related group work compared to their baseline impressions. They noted that more students appeared willing to take on roles, share ideas and interact around the robot, including students who were usually quiet or reluctant. Teachers also commented that students seemed more comfortable experimenting, making mistakes and asking for help even in other lessons. They reported that only two students continued to show persistent difficulties with participation and focus, despite support and adaptations.

5.2.1 Inclusion and Participation of Students with ASD (RQ1, RQ2)

Students' accounts show that participation became more possible when they were deliberately invited into the interaction and when expectations were made clear.

For the students with ASD in particular, the most positive experiences were often connected to concrete, hands-on tasks. Building, testing, placing sensors and observing the robot's behaviour provided accessible ways of participating that did not rely solely on fast-paced verbal discussion. Students described feeling included when others asked for their opinion, listened to their suggestions, or gave them a defined role within the group. One student explained, "*This time I felt people listened to me because they asked, 'What do you think the robot should do?' and then we tried my idea."*

These practical tasks appeared to offer a more comfortable and manageable form of engagement, while also allowing students to contribute visibly to the success of the group. Observations confirmed that these students often participated actively in building, testing and troubleshooting and in some cases took initiative in practical aspects of the work. Their engagement was especially noticeable when their actions produced an observable result in the robot's behaviour.

Recognition of contribution was also important. Students with ASD described positive moments when their ideas were followed or when their actions led to a visible improvement in the task, "*Normally I don't talk much, but with the robot I could show what I did instead of talking a lot and then people asked me how I did it."* This helped position them as competent participants rather than peripheral members of the group. At the same time, interview and observation data also point to important barriers. Students described difficulties related to the sensory and communicative demands of group work, especially during noisy or fast-moving moments. One student stated, "*I didn't like the noise…too many voices*." This captures a recurring issue observed during busier parts of the sessions, when overlapping talk and rapid shifts in attention made participation more difficult. In such moments, students with ASD sometimes withdrew from discussion, stepped back physically from the group, or focused narrowly on the robot instead of the wider collaborative interaction.

Teacher reflections emphasized that structured support was crucial in sustaining inclusion. Strategies such as assigning clear roles, establishing predictable routines, breaking tasks into smaller steps and allowing additional processing time were repeatedly described as important. One student noted feeling more comfortable when they knew "*exactly what my job is with the robot*," while another appreciated that "*the teacher told the others to wait so I can try*." Teachers also reported that when they adopted these strategies, benefits extended beyond students with ASD, supporting a more balanced and inclusive classroom environment overall.

5.3 Sustainability-Oriented Practices and Classroom Responsibility (RQ1, RQ3)

The second theme, Sustainability-Oriented Practices and Classroom Responsibility, captures how the robotics and AI activities became a context for enacting values related to fairness, care for shared resources, environmental awareness and inclusion. Across the data, sustainability was not treated as an isolated topic but was connected to everyday classroom practices such as sharing equipment, keeping the classroom tidy and designing robots that promoted responsible behaviour.

Students often described fairness in practical terms, especially in relation to taking turns, sharing materials and listening to one another's ideas. These concerns were closely linked to the use of common classroom resources, including laptops, robotics equipment and physical materials. One

student explained, "*I liked when our robot reminded us to share the laptops and then another time not waste paper. It felt like the robot was helping our classroom be fair and tidy*." This quotation illustrates how fairness and sustainability became intertwined within students' thinking. Inclusion was also connected to these classroom values. Students frequently described positive robot behaviour in terms of helping others, reminding peers respectfully and making sure "everyone" was included.

Sustainability appeared to become most meaningful when it was embedded in concrete, visible tasks. Students recalled scenarios involving recycling, saving energy, reducing waste and keeping the classroom clean, particularly when these ideas were built into the robot's actions, stories, or personalities. For example, a robot that "*does not like rubbish on the floor*" or reminds students not to waste paper transformed abstract environmental ideas into observable and interactive behaviour.

Observation data supported this as groups were seen designing and testing robots that reinforced classroom routines related to sharing materials, managing space and caring for the learning environment. These routines were social as well as environmental as students were often negotiating how to use limited resources fairly while simultaneously considering how their robot might encourage others to act responsibly. Teacher reflections further highlighted the value of this integration, suggesting that robotics activities provided a practical bridge between technology learning, social behaviour and environmental awareness.

At the same time, the data also indicate variation in how deeply different groups engaged with these issues. In some groups, students took up themes of fairness, care and sustainability in a thoughtful and visible way. In interviews students mentioned that programming the robot made them reflect on their own behaviour in the classroom, for example about sharing materials, including others in games, or not disturbing the lesson. In others, the focus remained more narrowly technical, with students concentrating on making the robot function without reflecting as much on the values embedded in its behavior.

Moreover, teachers identified several challenges in implementing the project. Time constraints were frequently mentioned, both in terms of preparing materials and in managing technical issues within limited lesson slots. They reported spending considerable time before and after lessons setting up robots, testing equipment and planning activities that integrated robotics and AI meaningfully with the curriculum. During lessons, technical breakdowns, such as connectivity problems, low batteries or malfunctioning sensors, sometimes disrupted the flow of activities and required rapid troubleshooting. These interruptions could lead to frustration for both students and teachers and occasionally reduced the time available for reflection or discussion. They, also, pointed to the need for ongoing professional development to feel confident in using sustainable practices and AI tools

while supporting students' questions about them. Some described feeling unsure about how to use sustainable practices in AI assisted educational robotics and blended learning systems in an age-appropriate way. Some worried that talking about surveillance or data collection might cause unnecessary anxiety, especially if students confused the classroom robot with more powerful technologies. Others felt uncertain about their own knowledge of AI ethics and did not want to provide incorrect or oversimplified information. This sometimes led them to focus more on concrete rules (e.g., "the robot should not spy") than on broader discussions about power, responsibility and bias.

5.4 Ethical Engagement with AI and Robotics in Practice (RQ3)

The third theme, Ethical Engagement with AI and Robotics in Practice, addresses how students and teachers interacted with AI-generated suggestions and how these interactions opened space for reflection on fairness, privacy, responsibility and care.

Students often described AI and educational robotics as "interesting", "helpful" or "funny", suggesting a generally positive orientation toward its use. However, their engagement was also evaluative. AI-generated suggestions were discussed, compared and revised rather than followed automatically. One student explained, "*The AI said the robot could tell the teacher if we were noisy, but we decided that was not fair. We changed it so the robot reminds everyone quietly instead."* This quotation is especially important because it shows students actively interpreting an AI suggestion through an ethical lens.

Some groups reported that they used an AI idea as a starting point and then adapted it, while others decided together to ignore a suggestion they did not like. A few students noted that AI sometimes helped them "*when we did not have ideas*" or "*when we were stuck*," and others commented that using AI made them think more about what was fair or kind behaviour for the robot "*The AI gave an idea that the robot should help kids who are often alone. I liked that because sometimes people forget about them, but the robot didn't."*

Students tended to value suggestions that were clear, realistic and relevant to their goals, while dismissing ideas they considered impractical or inappropriate. Some suggestions were seen as "too silly," too difficult," or "not realistic" for their classroom; while others functioned as useful starting points when groups felt uncertain about how to continue. For instance, they mentioned ideas that required resources they did not have, or that did not fit their understanding of what a classroom robot should do. In such cases, students either ignored the suggestion or modified it considerably. A small number of students also expressed unease about being observed. One student stated, "*I felt a bit strange when we talked about the robot watching us."* These concerns were usually reduced when

teachers clarified what the technology could and could not do and emphasised that students retained control over the robot's behaviour.

When asked whether the robot and AI helped the group work more fairly and respectfully, most students answered positively and gave examples. These examples included taking turns controlling the robot, listening to each person's idea before making a decision and using the robot to "help someone who needs it" in the classroom. Some students connected fairness to sustainability themes, for instance by mentioning that everyone should have access to the robot and that materials should not be wasted. They mentioned that the robot could remind others to include students who are often alone, or to make sure that everyone has a turn by emphasising that the robot should "help everyone" and not only certain classmates.

Observations further showed that ethical discussion often emerged naturally during the design process. As students debated what the robot should do, they also discussed what it should not do. Questions of fairness arose when students considered whether the robot should respond differently to different classmates, or whether it should prioritise one student's needs over those of the group. Privacy and surveillance became salient when ideas involved watching, reporting, or monitoring behaviour. Inclusion was also part of these discussions, especially when students considered whether the robot should "help everyone" or support classmates who might otherwise be left out.

Teacher reflections confirmed that the project created valuable opportunities to introduce ethical questions through concrete activities. They observed strong student interest in issues such as privacy and surveillance and considered robotics a useful way to make abstract concerns more understandable. They emphasised the importance of clear guidelines, age- appropriate resources and opportunities for professional learning to support them in facilitating these discussions responsibly. At the same time, they also reported uncertainty about how to explain more complex aspects of AI, such as data use, bias, or the functioning of AI systems, in developmentally appropriate ways. As a result, ethical guidance sometimes remained at a broad principle-based level, for example through rules such as "the robot should not spy," rather than extending into deeper discussion of why certain forms of monitoring or decision-making might be problematic.

6. Discussion

6.1 Discussion of results

Overall, students expressed clearly positive impressions of the robotics and AI project. For some, the project was memorable mainly because it involved technology and movement, which contrasted with more traditional, desk-based lessons. A number of students mentioned that the project motivated them to come to school or to participate more actively during the sessions. They described looking forward to the robotics lessons and feeling more engaged than in other subjects, especially when they could experiment, test their own ideas and see immediate feedback in the robot's behaviour. At the same time, a minority of students highlighted challenges and frustrations. Difficulties were most often related to technical problems or to time pressure when tasks felt too long or too complex for the available lesson time. These findings suggest that AI-assisted robotics can enhance engagement in blended primary classrooms, but that this engagement is contingent on reliable technological infrastructure and realistic task design rather than being an automatic effect of using robots or AI.

Therefore, AI-supported educational robotics in this blended primary classroom appeared to create new opportunities for collaboration and participation, particularly for students who usually take more peripheral roles in group work. This is visible when students describe "doing something important" for the group, take distinct roles such as coder, builder or tester and persist with debugging tasks even when these are challenging. These patterns suggest that the project supported not only cognitive engagement but also a sense of agency and belonging. This interpretation is reinforced by teachers' reflections that, over time, more students were willing to take on responsibilities, share ideas and ask for help during the blended sessions (Daniels, 2008).

At the same time, the findings suggest that these inclusive dynamics depended on intentional task design and structured scaffolding, rather than emerging automatically from the introduction of robots and AI. When roles were unclear or group composition was less carefully considered, some students monopolised the tablet or robot while others became passive observers, echoing concerns in previous work that ambitious robotics designs can clash with everyday classroom constraints and reproduce existing inequalities (OECD, 2024; Oleinikov et al., 2020). For students with ASD, participation was strongest when routines were predictable, instructions were concrete and sensory demands were managed. In noisy or rapidly changing situations, they tended to withdraw even when they were interested in the task, which resonates with findings that robotics can support participation for some autistic students but requires sensitive adaptation of the learning environment (Fachantidis et al., 2020; Ponticorvo et al., 2020). Viewed through the first analytic strand of interactional processes, the case shows that inclusive collaboration around robotics is fragile and contingent on design decisions

and classroom conditions. Furthermore, the findings challenge previous claims that robotics are inherently motivating by showing that students' enthusiasm declined when technical breakdowns interrupted activity flow or when tasks exceeded the available time.

The integration of simple, classroom-oriented AI tools functioned primarily as a mediating artefact that encouraged reflection. When AI proposed behaviours for a "helper" robot, for example, students not only evaluated whether suggestions were technically feasible but also whether they were kind or respectful. In several groups, pupils openly rejected AI ideas they perceived as unfair and collaboratively reformulated them, which supports the study's intention to frame AI as an object of critical engagement rather than a neutral support tool (De la Vall & Araya, 2023; Wang et al., 2024). These episodes also illustrate Vygotsky-inspired sociocultural ideas of mediation and scaffolding, with AI prompts, robots and teacher questions jointly structuring how students talk, negotiate and co-construct understandings of inclusion and responsibility (Vygotsky, 1978; Daniels, 2008). In this sense, the third theme on ethical engagement shows how AI can become part of the social and moral fabric of classroom interaction. By focusing on a blended design that combined online AI tools with face-to-face robotics work, the study adds new insight into how digital and physical spaces intersect to shape collaboration, ethical reflection, and participation.

Finally, the results highlight how sustainability-oriented practices emerged not only in explicit discussions but also in small-scale classroom routines. Sustainability and responsibility were embedded in everyday classroom practices, as students engaged with ideas such as fairness, resource sharing and care for materials through concrete and interactive scenarios, often linking these values to inclusive behaviours that promoted collective well-being. Ethical understanding similarly developed through hands-on decision-making, as students critically evaluated AI suggestions and considered what constituted fair and respectful technological behaviour, sometimes expressing discomfort with ideas such as surveillance. That was clear as students used the robot to support practices such as sharing materials, keeping the classroom tidy and remembering to recycle and they linked these behaviours to ideas of fairness and care. When they questioned AI-generated suggestions they were negotiating social sustainability, in line with arguments that sustainability in education is enacted through concrete relational practices rather than abstract meanings (Schina et al., 2020; Rumaiza, 2025). Interpreted through the second analytic strand, this suggests that educational robotics and simple AI can provide a positive start for social sustainability when they are embedded in everyday classroom concerns and routines.

The qualitative case study design, combined with a sociocultural lens, enabled a fined analysis of how inclusion, sustainability and ethical awareness were enacted in situated classroom interactions

rather than treated as abstract outcomes. By following one primary class through a four-week blended robotics and AI intervention, the study captured how participation and collaboration shifted across specific tasks, tools and classroom configurations. At the same time, the focus on a single case in a particular school context limits the extent to which the findings can be generalised, but strengthens their ecological validity by grounding the analysis in everyday teaching practices. The integration of student interviews, classroom observations and teacher reflections provided opportunities for triangulation, while also foregrounding possible tensions between adult and child perspectives on participation and responsibility.

Beyond the specific case, the findings speak to wider questions about what teachers and schools need in order to use AI-supported educational robotics in socially responsible ways. The case shows that inclusive collaboration, ethical reflection and sustainability-related practices did not simply "appear" because robots and AI tools were present; rather, they depended on teachers' ability to design structured roles, manage sensory and time constraints and frame AI as something to be questioned and negotiated. This implies that teacher education and in-service professional development need to move beyond basic technical training and explicitly address how to orchestrate group work, talk about fairness, privacy and sustainability with younger learners and critically adapt digital tools to local classroom realities. At a policy level, the study suggests that initiatives promoting AI or robotics in schools should be coupled with support for smaller class sizes, co-teaching or additional classroom support where possible, as well as time for collaborative planning. Taken together, these implications argue for viewing AI-supported educational robotics not as stand-alone "innovations" but as part of a broader policy commitment to inclusive and sustainable education (Shabani et al., 2010; González-Oliveras et al., 2025; Ryalat et al., 2025).

However, the aim of this research is transferability rather than broad generalisation. The single-case design enabled an in-depth exploration of interactional processes within the classroom context; however, it did not permit systematic comparison across different schools or classroom cultures. Similarly, although an experimental design may have allowed for stronger claims regarding the causal effects of the intervention, the decision to prioritise qualitative data foregrounded the situated perspectives and experiences of students and teachers. This approach, however, limited the extent to which changes in engagement or inclusion could be measured against control groups. Consequently, the findings are most likely to be transferable to primary classroom settings that share key characteristics with the case studied, such as a commitment to inclusive education and opportunities to integrate educational technologies rather than being uniformly applicable across all school contexts. Overall, the project demonstrates that robotics and AI can serve as powerful contexts for

fostering inclusive, responsible and reflective classroom communities but only when pedagogical design deliberately foregrounds these dimensions.

6.2 Connecting to previous research

Overall, the findings align with studies that present educational robotics as a context for collaboration, problem-solving and social skill development, while extending this work into a blended, socio-emotionally focused primary setting. Prior research has reported that robotics tasks can foster teamwork, communication and motivation across age groups (Guevara et al., 2025; Piedade et al., 2020; Tselegkaridis & Sapounidis, 2022) and that, for some students with ASD, robots can scaffold participation through concrete, multimodal interaction (Fachantidis et al., 2020; Schiavo et al., 2024). The present study echoes these patterns as students frequently described helping each other to debug, coordinate roles and manage time and students on the autism spectrum were observed taking active roles in building and testing when tasks were clearly structured and roles were stable. At the same time, this case adds to the literature by examining how inclusion is experienced in everyday group work and how these experiences evolve over a four-week period in a naturalistic classroom, rather than in short, researcher-controlled interventions (Alimisis, 2020; Pasalidou & Fachantidis, 2022).

The blended format also speaks to a gap between the robotics and blended learning literatures. Research on blended learning has emphasised how online components can extend reflection, provide flexible access and support self-paced work (Rasheed et al., 2020; Hong, 2024), while robotics studies tend to focus on face-to-face sessions (Silva et al., 2023; Edisherashvili et al., 2022). In this project, online tools were used for planning, experimenting with code and discussing AI suggestions and face-to-face sessions were dedicated to building, testing and negotiating group decisions. This configuration supports arguments that blended arrangements can reconfigure when and how students collaborate and negotiate roles (Kerimbayev et al., 2023), but it also confirms that without explicit structuring and teacher mediation, access to devices and voice can remain uneven, especially for students who are less confident with digital tools (OECD, 2024; Oleinikov et al., 2020). In this sense, the case contributes to education by illustrating how blended robotics can redistribute opportunities for participation.

Finally, the findings contribute to emerging debates on AI, sustainability and ethics in education. Much existing work highlights AI's potential to personalise learning and provide data-driven insights, while warning about risks related to privacy, bias and power (Carayannis & Morawska, 2022; Eden et al., 2024; Wang et al., 2024). This study shows that even modest, suggestion-based AI tools can make these issues tangible for primary students when they are embedded in concrete classroom scenarios and positioned as prompts for discussion. When pupils question whether it is "fair" for a

robot to prioritise certain peers, they are engaging with ideas linked to social sustainability and SDG-related concerns about equitable and safe learning environments (Schina et al., 2020; Lindberg & Haglind, 2021). In this way, the case illustrates how social sustainability can be traced in small-scale classroom practices, such as sharing materials, caring for the classroom and negotiating respectful robot behaviours, rather than only at policy level (Rumaiza, 2025; OECD, 2024). By foregrounding these micro-practices, the study extends current debates on AI in education towards a more situated, socio-technical understanding of sustainability.

Conclusion

This section summarises the main findings in relation to each research question. It brings together the main contributions of the study and reflects on their implications for AI-assisted educational robotics, inclusion and social sustainability in blended learning environments.

Overall, the study suggests that robotics and AI can act as valuable educational tools not only for developing technical skills but also for fostering inclusion, responsibility and ethical awareness in the classroom. Their impact, however, depends on thoughtful and intentional pedagogical design that makes these dimensions visible and meaningful for students. When supported in this way, such activities can contribute to more inclusive and reflective learning environments, where students engage with technology critically and collaboratively rather than passively.

RQ1: Students' social interaction, collaboration and sustainability related engagement

Overall, the integration of educational robotics in a blended format supported rich forms of social interaction and collaboration, while also creating concrete entry points for sustainability-related practices (Alimisis, 2020; Rasheed et al., 2020). Across sessions, students organised themselves into functional roles (such as builder, programmer and tester), negotiated turn-taking around the robots and devices and increasingly relied on peer explanation and joint problem-solving rather than waiting for the teacher, suggesting that the activities fostered both interdependence and a sense of shared responsibility for the group's progress (Guevara et al., 2025; Piedade et al., 2020; Tselegkaridis & Sapounidis, 2022). At the same time, the results show that patterns of participation remained sensitive to group composition, task clarity and sensory demands; when these were not carefully managed, some students, especially those with ASD or less confidence, were pushed to the margins, echoing earlier findings that everyday constraints and diverse needs shape how inclusive robotics work can become (Fachantidis et al., 2020; Oleinikov et al., 2020; Ponticorvo et al., 2020). In relation to sustainability, students not only talked about fairness and care in abstract terms but also enacted them in small, observable ways, for example by sharing materials, protecting the robots, or challenging AI-generated suggestions they perceived as unfair, indicating that social sustainability was being negotiated through everyday classroom practices rather than only in explicit discussions (Schina et al., 2020; Rumaiza, 2025).

RQ2: Teachers' perceptions of potential and limitations for inclusive, socially sustainable learning

Teachers perceived AI-supported educational robotics as a promising but demanding approach for fostering inclusive and socially sustainable learning environments, in line with broader debates on

human-centred, sustainable digitalisation in schools (OECD, 2024). In their reflective notes, they emphasised that the combination of hands-on robotics and blended organisation gave more students "a reason to participate," helped some quieter students with special educational needs to take visible roles and created natural occasions to talk about fairness, responsibility and care, echoing previous work on the social dimension of educational robotics and its potential for inclusion (González-Oliveras et al., 2025; Pnevmatikos et al., 2022; Schiavo et al., 2024). At the same time, they highlighted substantial constraints like time pressure, uneven prior experience with robotics, technical issues and the constant need to balance attention between groups made it challenging to provide the level of scaffolding they felt was needed for all students to benefit equally, reflecting tensions identified in earlier studies about inequitable access and classroom-level barriers to robotics integration (OECD, 2024; Oleinikov et al., 2020). Teachers also expressed uncertainty about how well prepared they were to respond to students' AI-related questions and to connect the robotics project with wider school priorities and resources, indicating that the potential they observed depends not only on thoughtful classroom design but also on ongoing support, professional development and favourable organisational conditions beyond the immediate project (Lindberg & Haglind, 2021; Eden et al., 2024; Wang et al., 2024).

RQ3: Participants' perceptions of ethical opportunities and risks in blended robotics activities

Students discussed whether robots should treat all classmates the same, whether certain AI suggestions felt "mean," "unfair" or "too strict," and whether it was acceptable for a robot to report individual behaviour to adults, demonstrating that even simple AI tools can trigger nuanced considerations of fairness, surveillance and responsibility when grounded in concrete scenarios (Schina et al., 2020; Lindberg & Haglind, 2021; Wang et al., 2024). Teachers, in turn, noted both positive moments, where AI outputs opened space for critical dialogue and points of concern, such as the risk that technology could unintentionally reinforce existing exclusions or that students might over-trust AI if its limitations were not continuously modelled and explained, concerns that resonate with wider discussions on power, data and vulnerability in AI-mediated education (Carayannis & Morawska, 2022; OECD, 2024). Together, these perceptions suggest that ethical awareness in this project emerged through ongoing negotiation between judgment, technological suggestions and classroom norms, underscoring the importance of treating AI-supported robotics as a socio-technical practice that requires active, reflective mediation rather than as a neutral add-on (Daniels, 2008; Allman, 2023 chap.12).

The thesis is therefore relevant not only for researchers but also for teacher education and policy discussions about digitalisation in schools. For teacher education, it provides concrete examples of

how blended robotics projects can be used to scaffold inclusive participation, ethical dialogue and sustainability-related practices in ordinary classrooms, which could be integrated into practicum seminars, design tasks or professional learning communities. For policymakers and school leaders, the study underlines that investments in AI-supported educational robotics should be accompanied by attention to equity, teacher workload and pupils' diverse needs. Without conditions that allow for careful planning, reflection and adaptation, there is a risk that such initiatives widen participation gaps rather than supporting the inclusive, socially sustainable education goals they often claim to serve (Carayannis & Morawska, 2022; OECD, 2024).

Limitations

The interpretive, single-case design of this study brings important limitations. The project took place in one primary school, with one class and a small group of teachers who already had some experience with educational robotics, over a four-week period. This aligns with recommendations for rich, bounded case studies that foreground context (Yin, 2018), but it means that the findings cannot be generalised to all schools, age groups or national settings. Instead, they offer what Yin (2018) calls "analytic generalisation," where readers judge the transferability of insights based on detailed descriptions of context, participants and activities. Other schools with different infrastructures, teacher expertise or student demographics might experience different patterns of collaboration, inclusion and ethical reflection (Mills et al., 2017)

A second limitation relates to the relatively short duration of the project and what can reasonably be inferred about longer-term change. The intervention took place over four weeks, which allowed patterns of collaboration, inclusion and ethical discussion to emerge, but it does not show whether these patterns would stabilise, deepen or fade over a full school year or across different topics. Some of the observed shifts, such as quieter students taking more active roles, or groups spontaneously questioning the fairness of certain robot behaviours, may partly reflect the novelty of the robotics project and the focused attention it received from teachers and the researcher. Without follow-up observations or comparison classes, it remains unclear to what extent the practices developed during the project would be integrated into everyday teaching or transferred to other subjects and activities (McGrath et al., 2019; Mills et al., 2017).

Finally, several practical constraints shaped the implementation. Technical issues, time pressure within the timetable and students' differing prior experience with robotics and online tools all influenced what could be achieved in the blended sessions. Shortened activities sometimes left less time for reflection and ethical discussion than originally planned and sensory factors such as noise and movement affected the participation of some students with ASD despite efforts to provide breaks

and stable roles (Ponticorvo et al., 2020; Schiavo et al., 2024). Moreover, the AI tools used were deliberately simple, did not process personal data and were tightly mediated by the teacher, which limits how far these findings can be applied to more complex AI systems such as adaptive learning platforms or learning analytics dashboards. The study therefore speaks most directly to low-stakes, suggestion-based AI supports embedded in teacher-designed activities.

Significance and future research

Despite these limitations, the study offers a theoretically and practically meaningful contribution by showing how AI-supported educational robotics can be used to promote collaboration, inclusion and ethical awareness in an ordinary primary classroom, rather than in highly controlled lab conditions. By framing educational robotics as a socio-technical practice embedded in everyday routines and by tracing social sustainability through concrete practices such as role distribution, resource sharing and fairness discussions, the thesis extends existing work on digitalisation and sustainability in education (Carayannis & Morawska, 2022; Schina et al., 2020; Rumaiza, 2025).

Beyond its contribution to research, the study also has implications for classroom practice, teacher education and policy. The findings show that inclusive collaboration, ethical dialogue and sustainability-related practices did not arise automatically from the presence of robots and AI, but depended on teachers' deliberate structuring of roles, careful management of classroom conditions and explicit framing of AI as something to be questioned rather than simply followed. This suggests a need for professional development that extends beyond technical skills to include the socio-emotional, ethical and organisational aspects of AI-supported robotics. In this way teachers will be able to design activities that make issues of fairness, participation and responsibility visible in everyday lessons. Teacher education programmes could, for example, draw on similar blended projects as design-based assignments, enabling future teachers to practise orchestrating group work, facilitating discussions about privacy and equity and adapting simple AI tools to diverse learners and local classroom realities. At a broader level, the case indicates that policy agendas promoting digitalisation, AI and robotics in schools are most likely to achieve their aims when they are coupled with conditions that allow for thoughtful implementation, such as time for collaborative planning, access to support and explicit guidance on inclusion and students' rights, rather than focusing solely on technology acquisition or performance outcomes (Carayannis & Morawska, 2022; Rumaiza, 2025).

Future research could build on this case by examining how similar AI-supported robotics projects unfold in different school contexts and over longer periods of time. Comparative studies across schools with varying resources, class sizes and inclusion policies could clarify which organisational

and pedagogical conditions most strongly support equitable participation, ethical dialogue and sustainability-oriented practices. In addition, involving teachers and students as co-designers of AIsupported activities would allow future studies to investigate how design choices and local policies shape both the opportunities and risks of integrating AI and robotics into everyday classroom life.

References

Adel, A. (2024). The convergence of intelligent tutoring, robotics and IoT in smart education for the transition from industry 4.0 to 5.0. *Smart Cities*, *7*(1), 325-369.

Alimisis, D. (2020). Emerging pedagogies in robotics education: towards a paradigm shift. In *Inclusive Robotics for a Better Society: Selected Papers from INBOTS Conference 2018, 16-18 October, 2018, Pisa, Italy* (pp. 123-130). Springer International Publishing.10.1007/978-3-030-24074-5_22

Atkinson, M. (2020). Sociocultural perspectives on learning. In Foundations of Learning and Instructional Design Technology. EdTech Books.

Bellas, F., Naya-Varela, M., Mallo, A. *et al.* Education in the AI era: a long-term classroom technology based on intelligent robotics. *Humanit Soc Sci Commun* 11, 1425 (2024). https://doi.org/10.1057/s41599-024-03953-y

Braun, V., & Clarke, V. (2022). *Thematic analysis: A practical guide.* Sage.

Bildik, C., & Altun, S. (2022). Undergraduates' conceptions of online education and technological mediation. European Journal of Open Education and E-learning Studies, 7(2), 173–190.

Carayannis, E. G., & Morawska-Jancelewicz, J. (2022). The futures of Europe: Society 5.0 and Industry 5.0 as driving forces of future universities. *Journal of the Knowledge Economy*, *13*(4), 3445-3471.

Daniels, H. (2008). Vygotsky and sociology. Routledge.

De la Vall, R. R. F., & Araya, F. G. (2023). Exploring the benefits and challenges of AI-language learning tools. *International Journal of Social Sciences and Humanities Invention*, *10*(01), 7569-7576.

Eden, C. A., Chisom, O. N., & Adeniyi, I. S. (2024). Integrating AI in education: Opportunities, challenges and ethical considerations. Magna Scientia Advanced Research and Reviews, 10(2), 006-013.

European Commission. (2021). *Ethics and data protection*. Brussels.

Fachantidis, N., Syriopoulou-Delli, C. K., & Zygopoulou, M. (2020). The effectiveness of socially assistive robotics in children with autism spectrum disorder. *International Journal of Developmental Disabilities, 66*(2), 113–121. https://doi.org/10.1080/20473869.2018.1495391

Fachantidis, N., Syriopoulou-Delli, C. K., Vezyrtzis, I., & Zygopoulou, M. (2020). Beneficial effects of robot-mediated class activities on a child with ASD and his typical classmates. *International Journal of Developmental Disabilities, 66*(3), 245–253. https://doi.org/10.1080/20473869.2019.1565725

Fachantidis, N., & Triantafyllidou, P. (2022). The impact of the social dimension of robots on school performance. Proceedings of the Hellenic Scientific Association for Information and Communication Technologies in Education Conferences, 460–467.

Fung, K. Y., Fung, K. C., Lui, T. L. R., Sin, K. F., Lee, L. H., Qu, H., & Song, S. (2025). Exploring the impact of robot interaction on learning engagement: a comparative study of two multi-modal robots. Smart Learning Environments, 12(1), 12.

Fortune Business Insights. (2026, February 16). *Educational robot market size, share and industry analysis by product, by education level and regional forecast 2026–2034*. https://www.fortunebusinessinsights.com/educational-robot-market-111582

Future Market Insights. (2025). *Educational robots market size & forecast 2025 to 2035*. Future Market Insights. https://www.futuremarketinsights.com/reports/educational-robots-market

Glava, E., & Seretis, K. (2022). Development of descriptive language through educational robotics in students with learning difficulties. Themes in Science and Technology in Education, 15, 55-66.

González-Oliveras, P., Engwall, O., & Wilde, A. (2025). Social Educational Robotics and Learning Analytics: A Scoping Review of an Emerging Field. *International Journal of Social Robotics*, 1-16.

Gubenko, A., Kirsch, C., Smilek, J. N., Lubart, T., & Houssemand, C. (2021). Educational robotics and robot creativity: an interdisciplinary dialogue. *Frontiers in Robotics and AI*, 178. doi.org/10.3389/frobt.2021.662030

Guevara, J., Leon, M., Vincent, J., Gunay, C., & Robertson, C. (2025, March). Project LegoLogic: Using Lego Spike Prime to Teach Essential Coding Concepts. In Society for Information Technology & Teacher Education International Conference (pp. 1883-1887). *Association for the Advancement of Computing in Education* (AACE).

Hong, L. (2024). The impact of educational robots on students' computational thinking: A meta-analysis of K-12. *Education and Information Technologies*, *29*(11), 13813-13838.

Kerimbayev, N., Nurym, N., Akramova, A., & Abdykarimova, S. (2023). Educational Robotics: Development of computational thinking in collaborative online learning. *Education and Information Technologies*, *28*(11), 14987-15009.

Edisherashvili, N., Saks, K., Pedaste, M. & Leijen, Ä. (2022). Supporting Self-Regulated Learning in Distance Learning Contexts at Higher Education Level: Systematic Literature Review. Front. Psychol. 12:792422. doi: 10.3389/fpsyg.2021.792422

Kouroupa, A., Laws, K. R., Irvine, K., Mengoni, S. E., Baird, A., & Sharma, S. (2022). The use of social robots with children and young people on the autism spectrum: A systematic review and meta-analysis. *PLoS ONE, 17*(6), e0269800.

Kurian, N. (2023). Toddlers and robots? The ethics of supporting young children with AI. *International Journal of Human Rights in Education, 4*(1)

Kurian, N. (2025). AI's empathy gap: The risks of conversational Artificial Intelligence for young children's well-being and key ethical considerations for early childhood education and care. *Contemporary Issues in Early Childhood*, *26*(1), 132-139.

Lemaignan, S., Daly, J., Newbutt, N., Rice, L., & Charisi, V. (2021). UNICEF Guidance on AI for Children: Application to the design of a social robot for and with autistic children. *arXiv:2108.12166.*

Lindberg, Y., & Haglind, T. (2021). The teacher as innovator in the pandemic: Changing practices and culture. *Tidsskriftet Læring Og Medier (LOM)*, *14*(24).

Allman, B. (2023). Sociocultural perspectives on learning. *Foundations of learning and instructional design technology (2nd ed.)*. May, T., & Perry, B. (2022). *Social research: Issues, methods and process*. McGraw-Hill Education (UK).

McGrath, C., Palmgren, P. J., & Liljedahl, M. (2019). Twelve tips for conducting qualitative research interviews. *Medical teacher*, *41*(9), 1002-1006.

Mills, G. E., Gay, L. R., & Airasian, P. (2017). *Educational research: Quantitative and qualitative methods – Applications.* Propobos.

National and Kapodistrian University of Athens. (n.d.). *Ethical guidelines for educational research involving children*. Department of Early Childhood Education. Retrieved May 22, 2025, from http://www.primedu.uoa.gr/fileadmin/primedu.uoa.gr/uploads/Pdfs/Grammateia/TELIKO_Kanones_deontologias_gia_tin_ekpaideytiki_ereyna_me_paidia.pdf

OECD. (2024). *The potential impact of artificial intelligence on equity and inclusion in education* (OECD Artificial Intelligence Papers, No. 23). OECD Publishing.

Oleinikov, V., Klimov, K., Rogachev, A., Kapytov, D., Budanov, V., Shtcherbov, R., ... & Buryak, A. (2020). The remote robotics education approach, its benefits and difficulties and the review of distance accessed robotics training ground. *Interactive Robotics Training Ground*, 1-11.

Pasalidou, C., & Fachantidis, N. (2022). Contextualizing educational robotics programming with augmented reality. In 2022 8th International Conference of the Immersive Learning Research Network (iLRN) (1–5). IEEE.

Pasalidou, C., Fachantidis, N., & Koiou, E. (2023). Using augmented reality and a social robot to teach geography in primary school. In M. Kurosu (Ed.), Human-Computer Interaction. Theoretical Approaches and Design Methods (371–385). Springer.https://doi.org/10.1007/978-3-031-05409-9_36

Pervin, N., & Mokhtar, M. (2022). The interpretivist research paradigm: A subjective notion of a social context. *International Journal of Academic Research in Progressive Education and Development*, *11*(2), 419-428.

Piedade, J., Dorotea, N., Pedro, A., & Matos, J. F. (2020). On teaching programming fundamentals and computational thinking with educational robotics: A didactic experience with pre-service teachers.*Education Sciences*,*10*(9), 214. https://doi.org/10.3390/educsci10090214

Pnevmatikos, D., Christodoulou, P., & Fachantidis, N. (2022). Designing a socially assistive robot for education through a participatory design approach: Pivotal principles for the developers. *International Journal of Social Robotics, 14*, 763–788. https://doi.org/10.1007/s12369-021-00826-1

Ponticorvo, M., Rubinacci, F., Marocco, D., Truglio, F., & Miglino, O. (2020). Educational robotics to foster and assess social relations in students' groups. Frontiers in Robotics and AI, 7, 78.

Rasheed, R. A., Kamsin, A., & Abdullah, N. A. (2020). Challenges in the online component of blended learning: A systematic review. *Computers & Education*,*144*, 103701. doi.org/10.1016/j.compedu.2019.103701

Rumaiza, S.M. (2025). Digital Innovations in Education Enhancing Sustainable Development Goals. In Digital Innovations for Renewable Energy and Conservation (pp. 75-98). IGI Global.

Ryalat, M., Almtireen, N., Al-refai, G., Elmoaqet, H., & Rawashdeh, N. (2025). Research and education in robotics: A comprehensive review, trends, challenges and future directions. *Journal of Sensor and Actuator Networks*, *14*(4), 76.

Shabani, K., Khatib, M., & Ebadi, S. (2010). Vygotsky's Zone of Proximal Development: Instructional implications and teachers' professional development. *English Language Teaching, 3*(4), 237–248. Canadian Center of Science and Education

Schiavo, F., Campitiello, L., Todino, M. D., & Di Tore, P. A. (2024). Educational robots, emotion recognition and ASD: New horizon in special education. Education Sciences, 14(3), 258.

Schina, D., Esteve-González, V., Usart, M., Lázaro-Cantabrana, J. L., & Gisbert, M. (2020). The integration of sustainable development goals in educational robotics: A teacher education experience. *Sustainability*, *12*(23), 10085.

Silva, R., Martins, F., Cravino, J., Martins, P., Costa, C., & Lopes, J. B. (2023). Using Educational Robotics in Pre-Service Teacher Training: Orchestration between an Exploration Guide and Teacher Role. *Education Sciences*,*13*(2), 210.

Singh, N., et al. (2022). Ethical considerations from child–robot interactions in under-resourced countries. *Frontiers in Robotics and AI, 9*, 867197.

Sorayyaei Azar, A., Elyas, T., Muthmainnah, M., & Curle, S. (Eds.). (2025). Innovations in Educational Robotics: Advancing AI for Sustainable Development: Advancing AI for Sustainable Development. IGI Global.

Stahl, N. A., & King, J. R. (2020). Understanding and using trustworthiness in qualitative research. *Journal of Developmental Education, 44*(1), 26–28.

Swedish Research Council. (2024). *Good research practice*. Swedish Research Council.

Tzagkaraki, E., Papadakis, S., & Kalogiannakis, M. (2021). Exploring the Use of Educational Robotics in primary school and its possible place in the curricula. In *Education in & with*

Robotics to Foster 21st-Century Skills: Proceedings of EDUROBOTICS 2020 (pp. 216-229). Cham: Springer International Publishing. doi: 10.1007/978-3-030-77022-8_19

Tselegkaridis, S., & Sapounidis, T. (2022). Exploring the Features of Educational Robotics and STEM Research in Primary Education: A Systematic Literature Review.*Education Sciences*,*12*(5), 305.

Velentza, A. M., Pliasa, S., & Fachantidis, N. (2021). Which one? Choosing favorite robot after different style storytelling and robots' conversation. *Frontiers in Robotics and AI, 8*, 700005. https://doi.org/10.3389/frobt.2021.70000

Verenikina, I. (2003). Understanding scaffolding and the ZPD in educational research. Proceedings of The Joint AARE/NZARE Conference.

Verenikina, I. (2003). Vygotsky's Zone of Proximal Development: Instructional implications and teachers' professional development. AARE Conference.

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes.* Harvard University Press.

Wan Hamedi, W. H., Awang Ali, F. D., Abdullah, W. Y., Ab Hamid, H., Shuhaimi, N. I. M., & Amir, M. M. (2025). AI as a digital scaffold: An integrative review of Vygotsky's Zone of Proximal Development in modern education. International Journal of Modern Education, 7(26), 579–593.

Wang, X., Pang, H., Wallace, M. P., Wang, Q., & Chen, W. (2024). Learners' perceived AI presences in AI-supported language learning: A study of AI as a humanized agent from community of inquiry. *Computer Assisted Language Learning*, *37*(4), 814-840.

Yin, R. K. (2018). *Case study research and applications: Design and methods* (6th ed.). Sage.

Younis, H. A., Ruhaiyem, N. I. R., Ghaban, W., Gazem, N. A., & Nasser, M. (2023). A systematic literature review on the applications of robots and natural language processing in education. Electronics, 12(13), 2864.

Appendices

Appendix 1: Researcher's Manuscript

This interview is about your experiences in the robotics and AI project. Your answers will help us understand how the activities worked and how we can improve them in the future.

There are no right or wrong answers. You can skip any question you do not want to answer. Your answers will be kept confidential and will only be used for research and improvement of teaching.

By continuing, you show that you agree to take part in this questionnaire. If you change your mind later, you can ask for your answers not to be used.

QUESTIONS

1. When you think about the robotics project we did in the blended/online sessions, what stands out most in your mind?
2. How did working with the robot in a group feel for you?
    - Can you describe a moment when collaboration worked well and
    - A moment when it was more difficult?
3. Did you feel included and listened to in your group during the activities?
    - What helped you feel included, or what made it harder?
4. We also used AI tools during the project. How did you experience working with AI together with the robot?
    - Were there moments when you helped someone else in your group with the robot or the code?
    - Were there moments when someone else helped you?
5. Can you remember any suggestion or answer from the AI that you especially liked or did not like and why?
6. In your opinion, did the robot and the AI help your group work respectfully with each other? Please give an example.

7. Some tasks tried to connect the project to sustainability (for example, caring for the environment or making the classroom more equal for everyone). What did you notice about that part?
8. Do you think using robotics and AI in this way is good for all students, including those who may be shy or have different learning needs? Why or why not?
9. Were there any moments when you felt worried or unsure about the robot or the AI?
10. Is there anything else you would like to tell us about your experience with the project, the robot, or the AI?

Appendix 2: Researcher's Manuscript

Autism-friendly adapted interview/ Researcher's Manuscript

This interview is about what you think and feel. There are no right or wrong answers. You can take your time. You can skip any question.

If you want help reading or writing, you can ask an adult you trust. Your answers are private and will only be used to improve teaching and for research.

1. Think about the robotics project we did (in class and online).
   What is the first thing you remember?
2. You worked with other students in a group and used the robot.
   Can you tell me about:
   - one time when working together felt good and
   - one time when working together was hard?
3. In your group, did you feel that other students listened to you?
   - What helped you feel listened to, or what made it difficult?
4. We also used an AI tool (a computer program that gives ideas and answers).
   How did it feel to use the AI together with the robot?
   - Did you ever help another student with the robot?
   - Did another student or the teacher help you?
5. Can you remember one idea or answer from the AI that you really liked or really did not like?

- What was it and why did you feel that way?

6. Do you think the robot and the AI helped your group to be fair and respectful?
   Please give an example if you can.

7. Some tasks were about sustainability. This means things like:

   - caring for the environment, or
   - making the classroom fair for everyone.
     What do you remember about this part?

8. Do you think using robotics and AI like this is good for all students, including students who are shy or learn in a different way? Why or why not?

9. Was there any time when you felt worried, stressed, or unsure about the robot or the AI?
   Please describe what happened.

10. Is there anything else you would like to tell us about your experience with the project, the robot, or the AI?